Exp No:16 Experiment to understand EDA-Quantitative and Qualitative analysis.

Description: Understand the importance of EDA-Quantitative and Qualitative analysis.

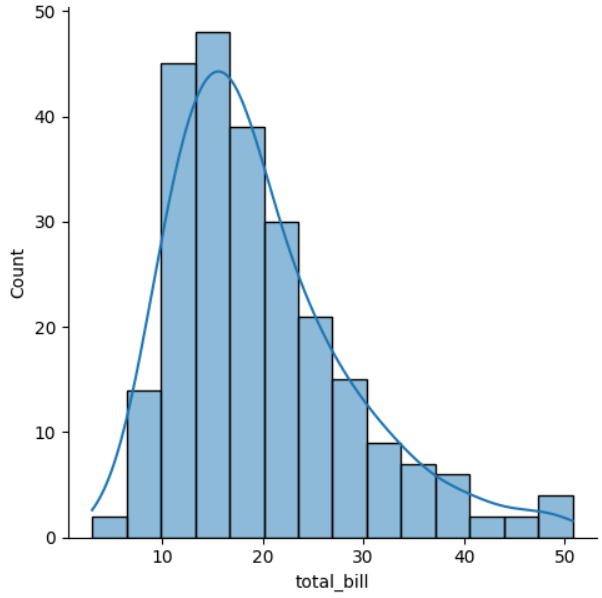
Code:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
tips=sns.load_dataset('tips')
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

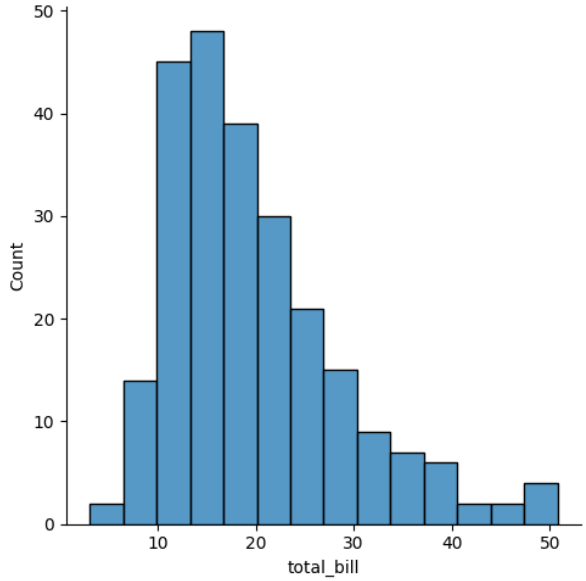
```
sns.displot(tips.total_bill,kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x7ce42ed0dd30>
```



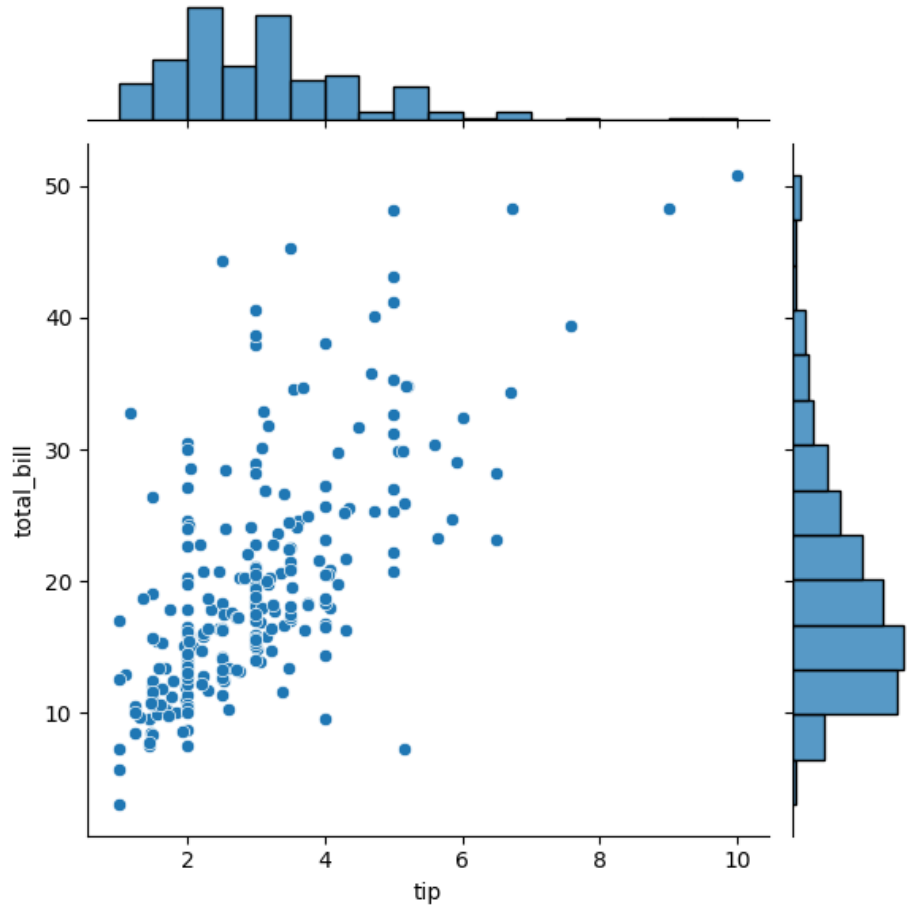
sns.displot(tips.total_bill,kde=False)

<seaborn.axisgrid.FacetGrid at 0x7ce42ead0910>



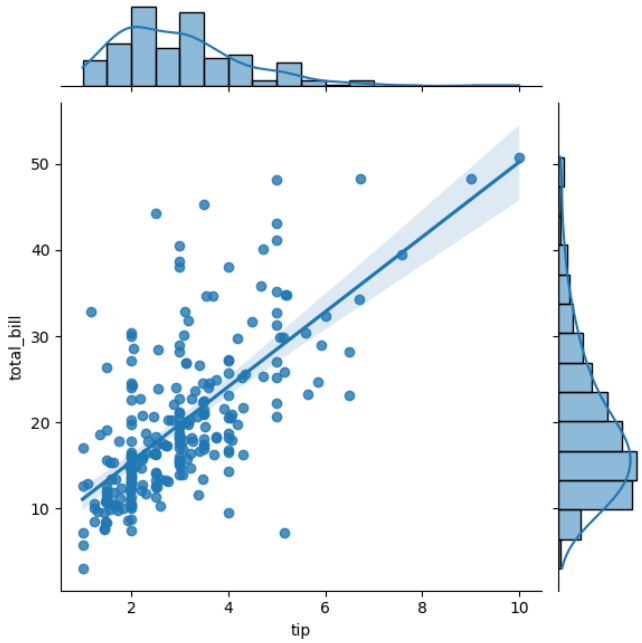
sns.jointplot(x=tips.tip,y=tips.total_bill)

<seaborn.axisgrid.JointGrid at 0x7ce42eb916a0>



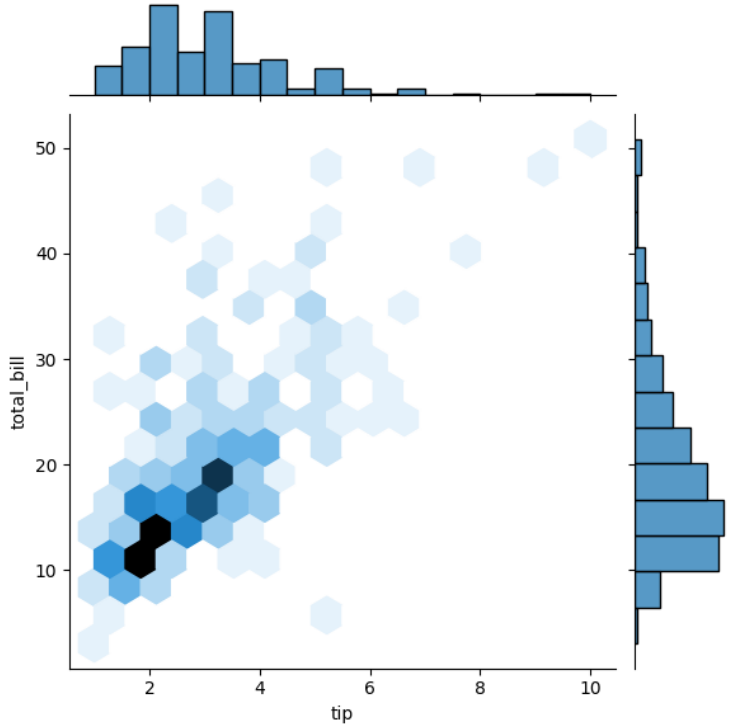
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")

<seaborn.axisgrid.JointGrid at 0x7ce42a89d950>



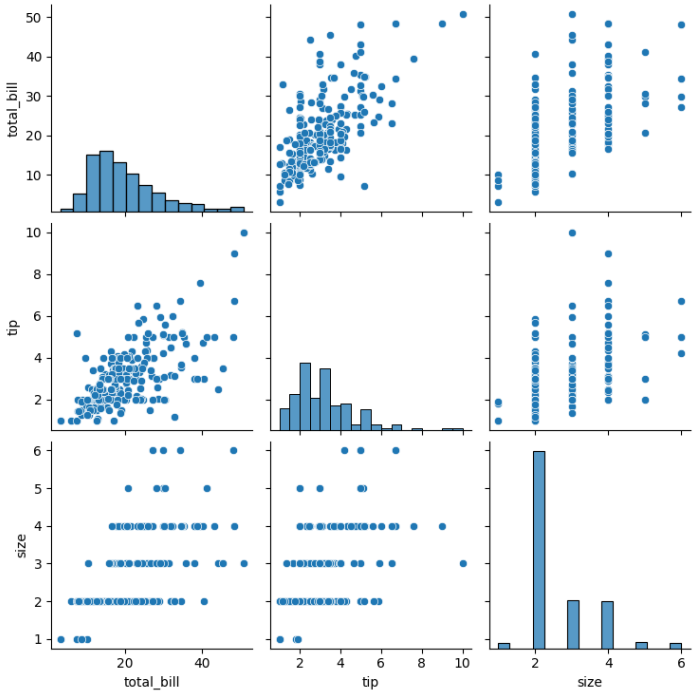
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")

<seaborn.axisgrid.JointGrid at 0x7ce42a7c3390>



sns.pairplot(tips)

<seaborn.axisgrid.PairGrid at 0x7ce42eb92900>



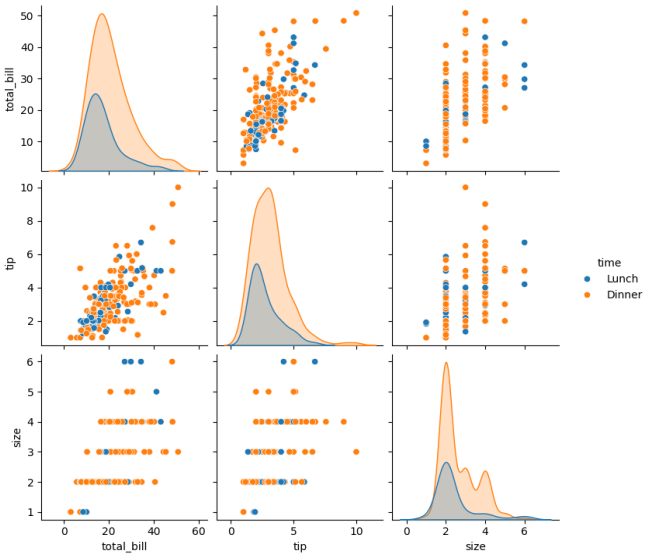
tips.time.value_counts()

	count
time	
Dinner	176
Lunch	68

dtype: int64

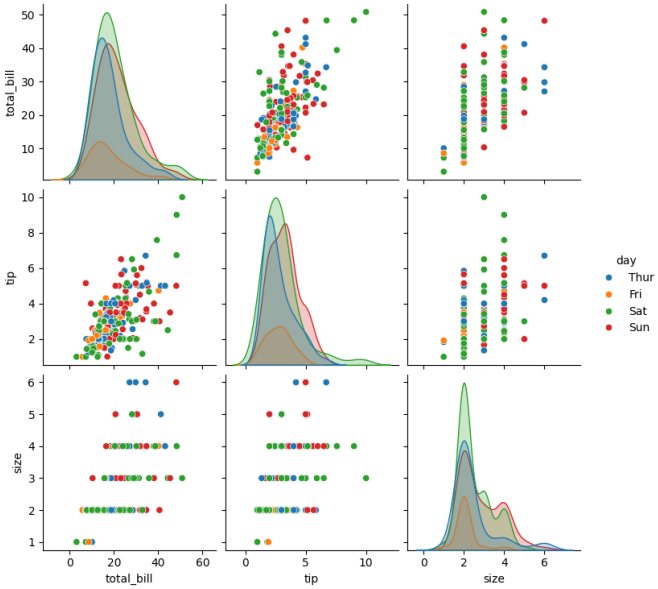
sns.pairplot(tips,hue='time')

<seaborn.axisgrid.PairGrid at 0x7ce42859ead0>



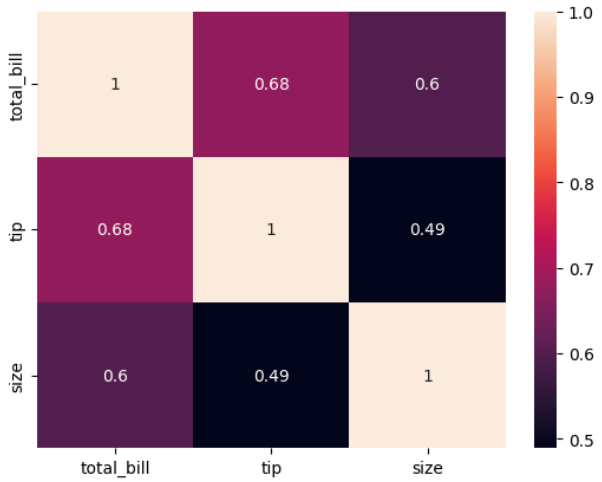
sns.pairplot(tips,hue='day')

<seaborn.axisgrid.PairGrid at 0x7ce423d83250>



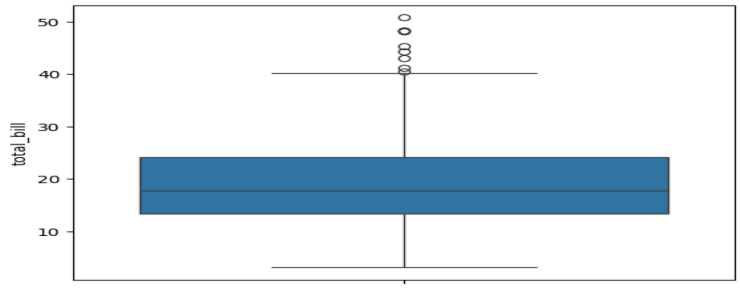
sns.heatmap(tips.corr(numeric_only=True),annot=True)

<Axes: >



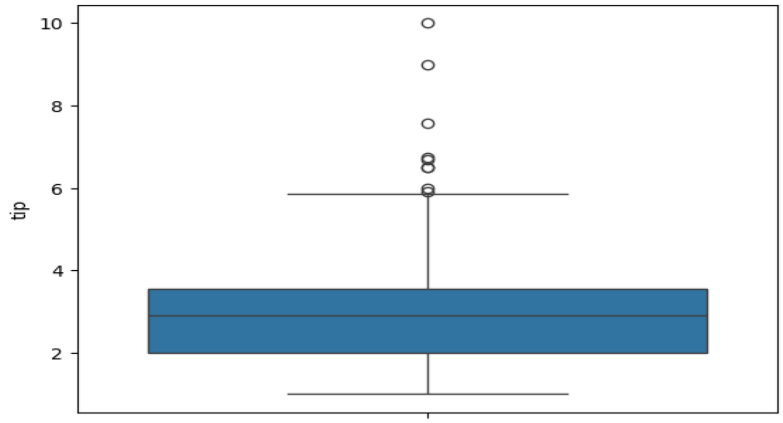
sns.boxplot(tips.total_bill)

<Axes: ylabel='total_bill'>



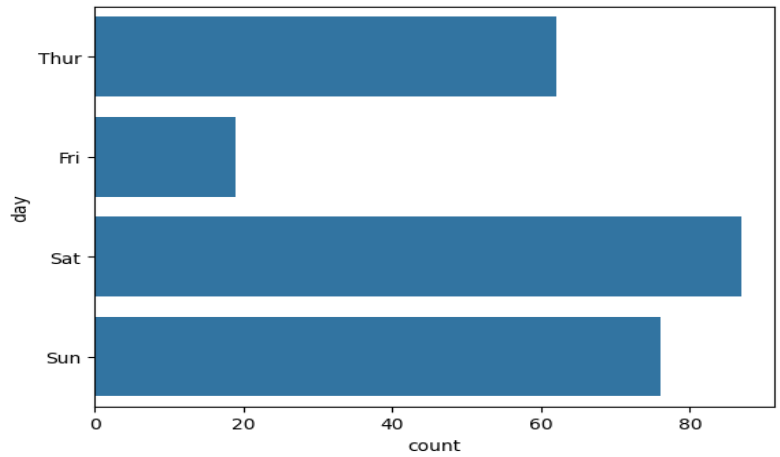
sns.boxplot(tips.tip)

<Axes: ylabel='tip'>



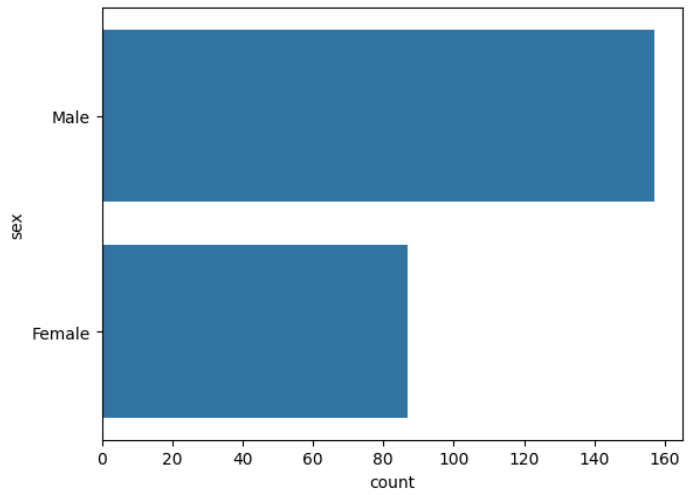
sns.countplot(tips.day)

<Axes: xlabel='count', ylabel='day'>



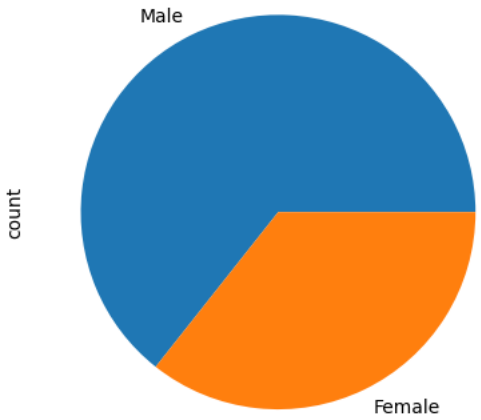
sns.countplot(tips.sex)

<Axes: xlabel='count', ylabel='sex'>



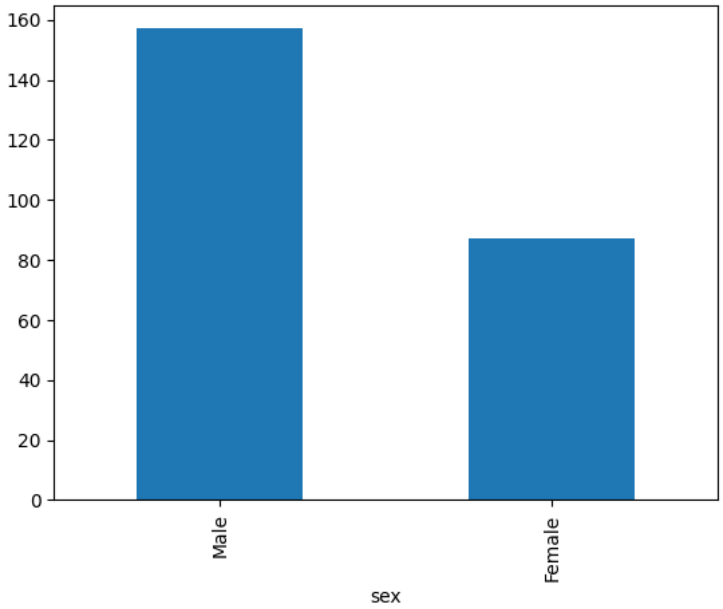
tips.sex.value_counts().plot(kind='pie')

<Axes: ylabel='count'>



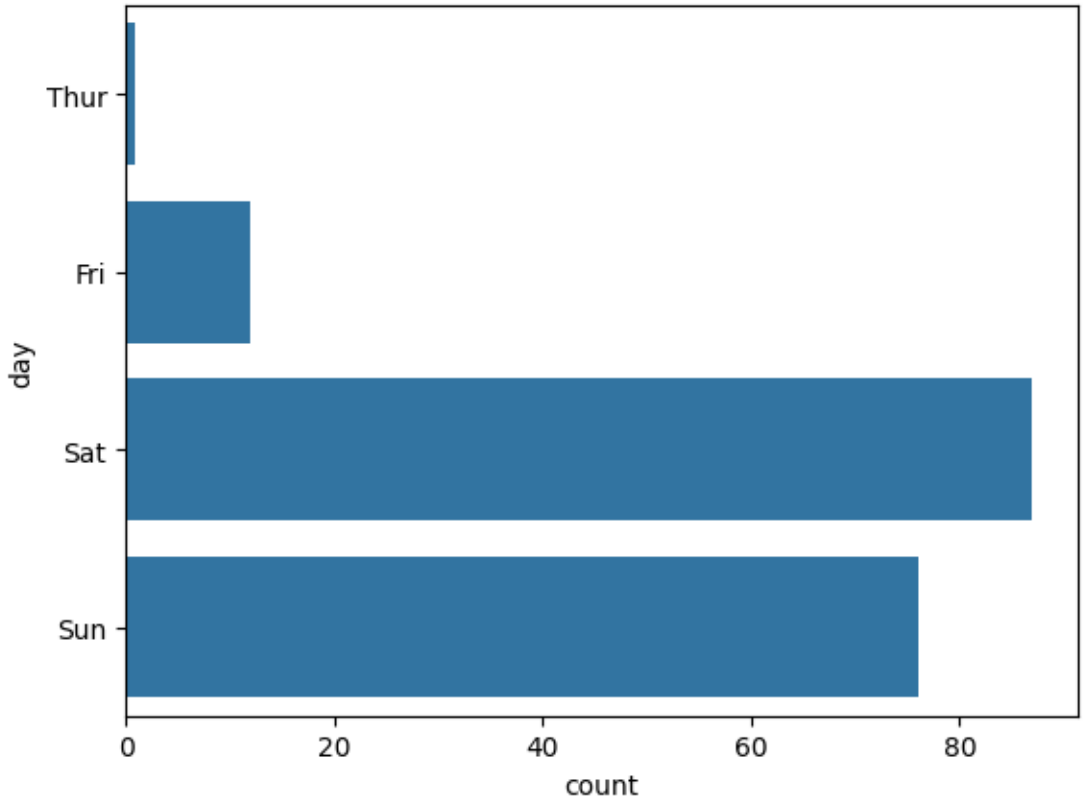
tips.sex.value_counts().plot(kind='bar')

<Axes: xlabel='sex'>



sns.countplot(tips[tips.time=='Dinner']['day'])

<Axes: xlabel='count', ylabel='day'>



Result:

Understood the importance of EDA-Quantitative and Qualitative analysis.